

Exercise Sheet 1 — Sample Answers

Introduction to Machine Learning Safety & Security

Exercise 1.1 — Safety vs. Security

■ *Question: Explain the difference between safety and security in the context of machine learning systems. Give one concrete example of a safety concern and one security concern for an ML-based autonomous vehicle.*

Answer:

Conceptual Distinction

Safety concerns the risk of unintended harm arising from the system's own behaviour — failures caused by design flaws, distribution shift, sensor noise, or unanticipated operating conditions. A safe system does not harm people or the environment *even when everyone is acting in good faith*.

Security concerns the risk of harm caused by deliberate, malicious interference from an adversary — someone who actively tries to manipulate inputs, steal model weights, or subvert the system's outputs for their own benefit.

In short: safety asks "what could go wrong by accident?", while security asks "what could an attacker make go wrong on purpose?". Both matter for ML systems but require different mitigations.

Concrete Examples for an Autonomous Vehicle

- **Safety concern — distribution shift:** The perception model was trained on clear daytime images. At dusk, or in heavy fog, image statistics differ substantially from the training distribution. The model may silently misclassify a pedestrian as background, causing the vehicle to fail to brake — not because of any attack, but because the model was never exposed to those conditions.
- **Security concern — adversarial patch:** An attacker places a carefully crafted sticker on a stop sign. The perturbation is imperceptible to humans but causes the neural network to classify the sign as a speed-limit sign, leading the vehicle to accelerate rather than stop. This is an intentional, targeted manipulation of the model's input.

Exercise 1.2 — The EU AI Act (optional)

■ *Question: 1. Name the main risk categories of the EU AI Act and briefly describe each. 2. Into which category would you place a perception system for (a) an autonomous vehicle and (b) a ChatBot? Justify your answer. 3. What obligations does the Act place on providers of systems in that category?*

Answer:

1. Main Risk Categories

Category	Description
Unacceptable Risk (Prohibited)	AI practices that pose a clear threat to fundamental rights (e.g. social scoring by governments, real-time biometric surveillance in public spaces). Banned outright.
High Risk	AI used in safety-critical infrastructure, education, employment, essential services, law enforcement, or administration of justice. Subject to strict requirements before market placement.
Limited Risk	Systems with transparency obligations only (e.g. chatbots must disclose they are AI). Lower regulatory burden.
Minimal / No Risk	Vast majority of AI (spam filters, recommendation engines). No specific obligations beyond general EU law.

2. Risk Classification of Perception Systems

a) Autonomous vehicle perception system — High Risk. Annex III of the EU AI Act explicitly lists AI systems used as safety components of vehicles covered by EU type-approval legislation (which includes road vehicles) as high-risk. A perception model that feeds directly into braking and steering decisions is exactly such a safety component: a failure could cause injury or death.

b) ChatBot — Limited Risk (presumed). A general-purpose conversational assistant does not fall into any Annex III category and therefore defaults to limited risk. The main obligation is transparency: users must be informed they are interacting with an AI. However, if the chatbot is deployed in a high-risk context (e.g. medical diagnosis support), it could be reclassified upward.

3. Obligations for High-Risk Providers (summary)

- Establish and maintain a quality-management system covering data governance, design documentation, and post-market monitoring.
- Keep technical documentation demonstrating conformity throughout the system's lifecycle.
- Implement a risk-management system that identifies, analyses, and mitigates foreseeable risks.
- Ensure training, validation, and test datasets are relevant, representative, and free from errors.
- Achieve a level of accuracy, robustness, and cybersecurity appropriate to the intended purpose.
- Provide human oversight measures so operators can intervene or halt the system.
- Register the system in the EU database before deployment and affix CE marking.
- Report serious incidents and malfunctions to national authorities.

Exercise 1.3 — Build the Right Model vs. Build the Model Right

■ *Question: Two complementary perspectives on ML quality can be summarised as: (1) "Build the right model" — correctness: does the model solve the intended problem? (2) "Build the model*

right" — robustness: is the model built and deployed with adequate care? For a pedestrian-detection model in an autonomous vehicle, describe one failure captured by each perspective.

Answer:

"Build the Right Model" — Correctness / Problem Validity

This perspective asks whether the model solves the *intended* problem — i.e. whether its specification and training objective correctly capture what the downstream system actually needs.

Example failure: The pedestrian-detection model is trained to maximise classification accuracy on a balanced dataset. The team treats it as a binary classifier that outputs "pedestrian present" or "not present". However, for the autopilot, what matters is the *probability* that a pedestrian occupies the vehicle's future path, weighted by distance and occlusion. Because the model was designed to answer a slightly different question (presence anywhere in the frame vs. in the ego-lane), it correctly identifies pedestrians on pavements but fails to flag the cases the autopilot actually needs to act on. This is a correctness failure: the right model for the task was never built.

"Build the Model Right" — Robustness / Engineering Quality

This perspective asks whether the model was developed and deployed with adequate care: sound data pipelines, appropriate evaluation, documentation, and operational monitoring.

Example failure: The model achieves high accuracy on the held-out test set, but the test set was sampled from the same clear-weather CARLA sessions as the training data. No separate evaluation under rain, night, or sensor-noise conditions was performed, and confidence calibration was never checked. In deployment the model produces high-confidence predictions on out-of-distribution inputs without any warning signal. This is a robustness / engineering failure: the model was not built or validated with sufficient care.

Exercise 1.4 — First Impressions (Safety Case Evidence)

■ *Question: Before looking at any data or model, think about the CARLA perception system described above. 1. Name three situations in which you would not trust this model's predictions. Explain why. 2. Which of the three output labels (traffic light / pedestrian / vehicle present) do you consider most safety-critical if missed? Justify your choice. 3. What additional information would you want before deploying this model in a real vehicle?*

Answer:

1. Three Situations Where Predictions Should Not Be Trusted

- **Adverse weather (rain, fog, snow):** The training data is exclusively clear/sunny. In rain, lenses produce halos and streaks; in fog, contrast drops sharply; in snow, lane markers and even large vehicles can disappear. The model has never seen these statistics and will produce unreliable outputs, potentially with unwarranted high confidence.
- **Night-time or low-light conditions:** Daytime-only training means the model has no experience with headlight glare, deep shadows, or the altered colour distribution of artificial

lighting. Pedestrians wearing dark clothing at night are likely to be missed, which is precisely the scenario where correct detection matters most.

- **Unusual or rare object appearances (edge cases):** CARLA provides a limited set of 3-D asset types and textures. Real-world pedestrians wear costumes, carry large objects, travel in wheelchairs, or ride unusual vehicles. Vehicles include emergency services, large lorries, and motorcycles with unusual proportions. These appear far outside the CARLA-asset distribution, causing potential misclassification.

2. Most Safety-Critical Label

Pedestrian present is the most safety-critical label to miss (false negative).

Pedestrians are uniquely vulnerable: unlike other vehicles, they offer no protective structure, and a collision at even moderate speed is likely to cause death or severe injury. Furthermore, pedestrian behaviour is less predictable than vehicle behaviour — they may step into the road suddenly. Missed traffic lights can cause collisions with vehicles, which is also serious; however, modern vehicles provide some mutual protection and traffic patterns are more predictable. A missed pedestrian with no other safeguard in the pipeline represents the highest irreversible harm potential.

3. Additional Information Required Before Deployment

- **Performance breakdown by condition:** Accuracy, precision, recall, and F1 disaggregated by weather, lighting, distance, and object size — not just aggregate accuracy.
- **Confidence calibration data:** Is the model's confidence score a reliable probability? Are high-confidence errors possible on OOD inputs?
- **Failure mode analysis:** What are the most common false-negative scenarios, and is there a pattern (e.g. specific pedestrian poses or distances)?
- **Downstream integration specification:** Exactly how does the autopilot consume the binary outputs — thresholds, fusion logic, fallback behaviour?
- **Real-world vs. CARLA domain gap analysis:** Quantitative comparison of image statistics (colour histograms, texture distributions) between CARLA renders and real camera footage.
- **Operational design domain (ODD):** Formal specification of the conditions under which the system is intended to operate, so the gap between ODD and training distribution is explicit.

Exercise 1.5 — Course Topics and the Safety Case (optional)

■ *Question: This seminar covers: testing and validation, explainability, uncertainty estimation, anomaly detection, adversarial machine learning, and alignment. For each topic, briefly explain how it could contribute evidence to the safety case for the CARLA model.*

Answer:

Testing and Validation

Provides direct empirical evidence of model performance within and outside the training distribution. For the CARLA model, structured test suites covering night, rain, and edge-case assets generate quantitative pass/fail evidence that the model meets minimum recall requirements for pedestrian

detection — a core safety claim.

Explainability

Supports the claim that the model is making predictions for the right reasons. Saliency maps or concept activation vectors can show whether the pedestrian classifier attends to human body features rather than spurious correlations (e.g. background textures common in CARLA). This provides evidence against shortcut learning.

Uncertainty Estimation

Enables the system to signal when it is operating outside its competence, which is critical for a model trained only on clear daytime data. Calibrated uncertainty estimates (e.g. from Monte-Carlo Dropout or ensembles) allow the autopilot to fall back to a safe state rather than act on an unreliable prediction. This constitutes evidence for safe degradation under distribution shift.

Anomaly Detection

Provides a runtime guard that can detect when an input image is sufficiently different from the training distribution to make model outputs untrustworthy. For the CARLA model, an anomaly detector trained on clear-weather features would flag night or rain images before the classifier is applied, supplying evidence that the system has a defined and monitored operational envelope.

Adversarial Machine Learning

Assesses the system's robustness against deliberate manipulation of camera inputs — relevant both as a security concern and, indirectly, as a safety concern (adversarial examples overlap with naturally occurring hard samples). Adversarial evaluation results and documented certified radii contribute evidence to the security sub-case of the overall safety argument.

Alignment

Ensures that the objectives the model optimises are coherent with the true goals of the deployment context. For the CARLA perception system, alignment considerations motivate the choice of loss function (e.g. asymmetric cost of false negatives vs. false positives for pedestrians) and the specification of what the classifier should and should not respond to. This supports the "build the right model" dimension of the safety case.

These answers are provided for study and discussion purposes. References to the EU AI Act reflect the regulation as of early 2025; article numbering may shift as implementing acts are adopted.